Student: _____ **Date:** _____	**Instructor:** Javier Arsuaga **Course:** MAT 021B E01-E07 WQ 2021	**Assignment:** Final Exam

(1) I pledge that I will neither give nor receive any aid from any other person during this assignment, and that the work presented here is entirely my own. (2) Problem #7 is extra-credit (3) **SHOW ALL YOUR WORK (and explain your reasoning) in all your problems to get full credit** (4) Any problem with no explanation or work will receive at most 10% of the total credit for the problem

Signature _____ Date _____

1. For the function given below, find a formula for the Riemann sum obtained by dividing the interval [0,12] into n equal subintervals and using the right-hand endpoint for each c_k. Then take a limit of this sum as $n \to \infty$ to calculate the area under the curve over [0,12].

   $f(x) = 144 - x^2$

   Write a formula for a Riemann sum for the function $f(x) = 144 - x^2$ over the interval [0,12].

   $S_n =$ _____ (Type an expression using n as the variable.)

   The area under the curve over [0,12] is _____ square units.
   (Simplify your answer.)

2. A solid lies between planes perpendicular to the x-axis at $x = -12$ and $x = 12$. The cross-sections perpendicular to the x-axis between these planes are squares whose bases run from the semicircle $y = -\sqrt{144 - x^2}$ to the semicircle $y = \sqrt{144 - x^2}$. Find the volume of the solid.

   The volume of the solid is _____ cubic units.
   (Simplify your answer.)

3. Evaluate the integral using integration by parts.

   $$\int e^{5x} \cos(8x)\, dx$$

   $\int e^{5x} \cos(8x)\, dx =$ _____

4. Evaluate $\int_3^{11} \frac{4}{s^2} ds$ using the trapezoidal rule and Simpson's rule. Determine

i. the value of the integral directly.
ii. the trapezoidal rule estimate for $n = 4$.
iii. an upper bound for $|E_T|$.
iv. the upper bound for $|E_T|$ as a percentage of the integral's true value.
v. the Simpson's rule estimate for $n = 4$.
vi. an upper bound for $|E_S|$.
vii. the upper bound for $|E_S|$ as a percentage of the integral's true value.

The value of $\int_3^{11} \frac{4}{s^2} ds$ is _____.
(Round to four decimal places as needed.)

The trapezoidal rule estimate of $\int_3^{11} \frac{4}{s^2} ds$ for $n = 4$ is _____.
(Round to four decimal places as needed.)

The upper bound on $|E_T|$ is _____.
(Round to four decimal places as needed.)

The upper bound for $|E_T|$ as a percentage of the integral's true value is _____.
(Round to four decimal places as needed.)

The Simpson's rule estimate of $\int_3^{11} \frac{4}{s^2} ds$ for $n = 4$ is _____.
(Round to four decimal places as needed.)

The upper bound on $|E_S|$ is _____.
(Round to four decimal places as needed.)

The upper bound for $|E_S|$ as a percentage of the integral's true value is _____.
(Round to four decimal places as needed.)

5. Perform a test to detemine if $\int_0^{\ln 5} 6x^{-2} e^{-6/x} dx$ converges.

$\int_0^{\ln 5} 6x^{-2} e^{-6/x} dx =$ _____
(Type an exact answer.)

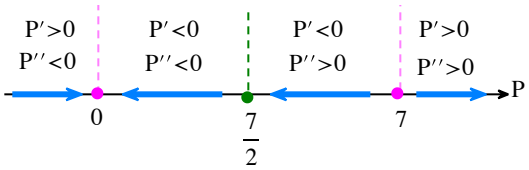
Based on the test, does $\int_0^{\ln 5} 6x^{-2} e^{-6/x} dx$ converge or diverge?

- Diverge
- Converge

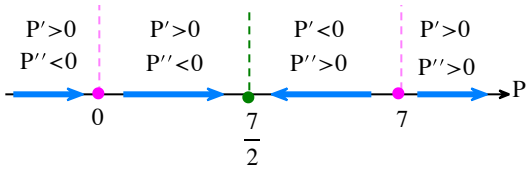
6. The autonomous differential equation $\frac{dP}{dt} = 2P(P-7)$ represents a model for population growth. Use a phase line analysis to sketch solution curves for P(t), selecting different starting values P(0). Which equilibria are stable, and which are unstable?

Construct a phase line. Choose the correct answer below.

A.



B.



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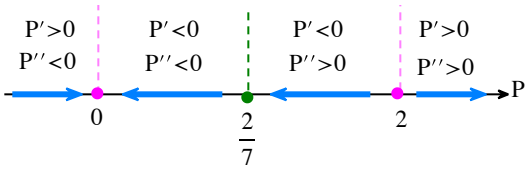

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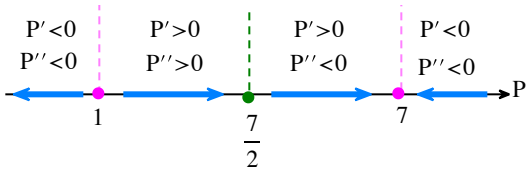
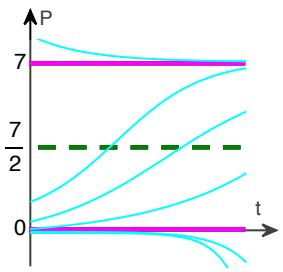
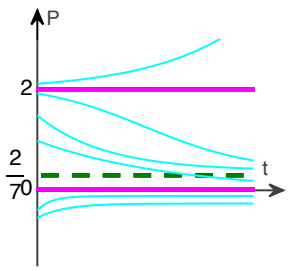


Which graph below correctly sketches the solution curves for $\frac{dP}{dt} = 2P(P-7)$?

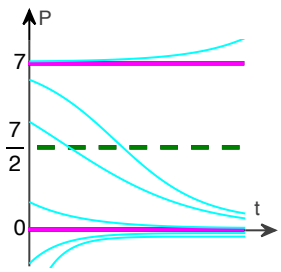
A.



B.



C.



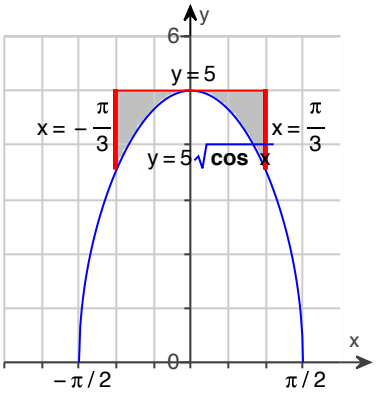
Which equilibria are stable?

- **A.** The equilibrium value(s) _____ is(are) stable. (Use a comma to separate answers as needed. Type an integer or a fraction.)
- **B.** None of the equilibria are stable.

Which equilibria are unstable?

- **A.** The equilibrium value(s) _____ is(are) unstable. (Use a comma to separate answers as needed. Type an integer or a fraction.)
- **B.** None of the equilibria are unstable.

7. Use the washer method to find the volume of the solid generated by revolving the shaded region about the x-axis.



V = _____

(Type an exact answer, using π as needed.)